Indian Institute of Technology Jodhpur
MAL1020
Tutorial Sheet 5

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1. Let $C[a, b]$ be the vector space of all real-valued continuous functions defined on the closed interval $[a, b]$. We define

$$\langle f, g \rangle = \int_a^b f(t)g(t)\,dt, \quad \text{for all } f, g \in C[a, b].$$

   Show that $C[a, b]$, equipped with the above operation, is an inner product space.

2. Let $\{v_1, v_2, \ldots, v_n\}$ be a spanning set for an inner product space V. If $v \in V$ satisfies

$$\langle v, v_i \rangle = 0, \quad \text{for each } i = 1, 2, \ldots, n,$$

   show that $v = 0$.

3. The three vectors $v_1 = [1, 2, 1]^T$, $v_2 = [2, 1, -4]^T$, $v_3 = [3, -2, 1]^T$ are mutually orthogonal. Express the vector $v = [7, 1, 9]^T$ as a linear combination of v_1, v_2, v_3.

4. Verify whether the given function $\langle u, v \rangle$ defines an inner product on $\mathbb{R}^n$. If it is an inner product, check if the given vectors are orthogonal.
   Given the inner product:

$$\langle u, v \rangle = u_1v_1 + 2u_2v_2 + u_3v_3$$

   where $u, v \in \mathbb{R}^3$, determine if this satisfies the inner product properties (linearity, symmetry, and positive definiteness).
   Then, check if the vectors

$$u = (1, -1, 2), \quad v = (2, 1, 0)$$

   are orthogonal under this inner product.

5. Let A be a 3×3 matrix whose columns represent a set of vectors in $\mathbb{R}^3$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$$

   (a) Apply the Gram-Schmidt process to find an orthonormal basis for the column space of A.
   (b) Verify that the obtained basis vectors are orthogonal.

6. Let $V = \mathbb{R}^3$ with the inner product defined as

$$\langle u, v \rangle = 3u_1v_1 + 2u_2v_2 + u_3v_3.$$

(a) Verify whether this inner product satisfies the properties of an inner product.

(b) Find the norm of the vector $u = (1, 2, -1)$ under this inner product.

(c) Compute the angle between the vectors $u = (1, 2, -1)$ and $v = (2, -1, 3)$ using this inner product.

7. Use Gram-Schmidt process to convert the given basis B of the Euclidean space $\mathbb{R}^3$ with standard inner product into an orthogonal basis.

   (a). $B = \{(1, 2, -2), (2, 0, 1), (1, 1, 0)\}$;

   (b). $B = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$.

8. Apply Gram-Schmidt process to find an orthonormal basis for the Euclidean space $\mathbb{R}^3$ with standard inner product, that contains

   (a). the vector $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0\right)$;

   (b). the vectors $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}\right), \left(\frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$.

9. Consider the inner product space of continuous functions on the interval $[0, 1]$ with the inner product defined as:

$$\langle f, g \rangle = \int_0^1 f(x)g(x)\,dx.$$

   Given the functions:

$$f_1(x) = x, \quad f_2(x) = x^2 - \frac{1}{3}, \quad f_3(x) = x^3 - \frac{3}{5}x,$$

   answer the following:

   (a) Compute the inner products $\langle f_1, f_2 \rangle$, $\langle f_1, f_3 \rangle$, and $\langle f_2, f_3 \rangle$.

   (b) Determine whether the set $\{f_1, f_2, f_3\}$ forms an orthogonal set.

   (c) If they are not orthogonal, apply the Gram-Schmidt process to construct an orthogonal basis for the space spanned by $\{f_1, f_2, f_3\}$.

10. Apply Gram-Schmidt process to obtain an orthonormal basis of the subspace of the Euclidean space $\mathbb{R}^4$ with standard inner product, spanned by the vectors

    (a). $(1, 1, 0, 1), (1, -2, 0, 0), (1, 0, -1, 2)$;

    (b). $(1, 1, 1, 1), (1, 1, -1, -1), (1, 2, 0, 2)$.